

IMNM

NECROSIS + MINIMAL INFLAMMATION

STOPPING ≠ IMPROVEMENT

STATIN

IVIG MONOTHERAPY

YOUNG = mimics LGMD

Cancer screening card

MHC-I ↑

If Ab+ → biopsy NOT needed

DO NOT DELAY

SEVERE

AGGRESSIVE

REFRACTORY

IMNM: necrosis CK >10x HMGR = IVIG SRP = aggressive

ASyS

LUNGS DECIDE PROGNOSIS

PFT ↓ FVC ↓ DLCO

JHE

Triphasic white-blue-red-cle-red

RAYNAUD

MECHANIC'S HANDS

LUNGS DECIDE PROGNOSIS

ILD DOMINATES

Perimysial fragmentation and perifascicular muscle fiber necrosis

DM = atrophy VS ASyS = NECROSIS

HLA-DR stain

IFN-γ driven

Alk Phos +

PRED MMF AZA RTX

DRUG OF CHOICE for ILD

AZA TX AVOID in ILD

Jo-1

PL-7

PL-12

EJ

KS

HISTIDYL

MOST COMMON

25-35% of myositis

AVOID in ILD

1. CK >10x (IMNM)

WHAT YOU SEE

The CK thermometer top-left with its mercury blown past the '>10x normal' mark — the maxed-out thermometer shows IMNM produces the highest CK of all the myopathies.

WHAT IT ENCODES

Immune-mediated necrotizing myopathy (IMNM) produces the HIGHEST creatine kinase of all the inflammatory myopathies — frequently >10x (and often 10–50x) the upper limit of normal — together with rapidly progressive, severe proximal weakness. The CK closely tracks disease activity and is used to follow response. The two defining serologies are anti-SRP and anti-HMGCR.

BOARD TRAP

WRONG answer: seeing a very high CK with proximal weakness and labeling it polymyositis. Discriminator: a CK >10x ULN with rapidly worsening weakness and a NECROTIZING biopsy (myofiber necrosis, scant inflammation) is IMNM, not PM — confirm with anti-SRP/anti-HMGCR antibodies.

2. Necrosis + minimal inflammation

WHAT YOU SEE

The forge-and-anvil scene labelled 'NECROSIS + MINIMAL INFLAMMATION' — the muscle being hammered/destroyed with almost no inflammatory crowd is the IMNM biopsy picture: dead fibers, sparse infiltrate.

WHAT IT ENCODES

The biopsy signature of IMNM is scattered MYOFIBER NECROSIS and regeneration with SPARSE or absent inflammatory cell infiltrate — distinctly different from PM (endomysial CD8+ T cells invading non-necrotic fibers) and DM (perifascicular atrophy). Complement deposition (MAC) on the sarcolemma of non-necrotic fibers and MHC-1 upregulation are seen. The relative lack of inflammation is the key clue.

BOARD TRAP

WRONG answer: dismissing the diagnosis of an inflammatory myopathy because the biopsy shows little inflammation. Discriminator: IMNM is an autoimmune, immune-mediated myopathy DESPITE minimal inflammatory infiltrate — necrosis with few inflammatory cells plus high CK and anti-SRP/HMGCR is the pattern, and it still requires immunosuppression.

3. Statin trigger

WHAT YOU SEE

The pill bottle stamped 'STATIN' on the forge bench — the statin is the trigger drug for anti-HMGCR necrotizing myopathy (HMG-CoA reductase is its target).

WHAT IT ENCODES

Statin exposure can trigger anti-HMGCR-positive IMNM (HMG-CoA reductase is the statin target). Unlike self-limited statin myalgia/myopathy, this is an AUTOIMMUNE disease that PERSISTS or worsens after the statin is stopped and requires immunosuppression (steroids + IVIG ± methotrexate/rituximab). It can also occur without statin exposure, especially in younger patients.

BOARD TRAP

WRONG answer: stopping the statin and reassuring the patient that the weakness will resolve. Discriminator: simple statin toxicity improves within weeks of withdrawal with a falling CK; anti-HMGCR IMNM KEEPS PROGRESSING after statin cessation and needs aggressive immunosuppression — persistence after stopping the drug is the discriminator.

4. MHC-1 up

WHAT YOU SEE

The 'MHC-1' upregulation label on the left wall — drawn as a shared marker tag, it flags an immune-mediated myopathy versus a non-inflammatory dystrophy, but does NOT pick a subtype.

WHAT IT ENCODES

Sarcolemmal MHC class I (MHC-1) upregulation on the surface of muscle fibers is a SHARED biopsy marker across inflammatory myopathies (PM, DM, IMNM, IBM) and helps distinguish autoimmune myositis from non-inflammatory dystrophies, where MHC-1 is typically absent. Normal muscle fibers do not express MHC-1; its induction signals an immune-mediated process.

BOARD TRAP

WRONG answer: using MHC-1 upregulation to pin a specific subtype. Discriminator: MHC-1 is NON-specific among the inflammatory myopathies — it confirms an immune-mediated myopathy versus a dystrophy but does not differentiate PM from DM from IMNM; the infiltrate pattern (CD8 endomysial vs perifascicular vs necrosis) makes that call.

5. If Ab+ biopsy not needed

WHAT YOU SEE

The note reading 'If Ab+ → biopsy NOT needed' on the lower-left wall — a positive disease-specific antibody in a classic picture lets you skip the biopsy and start treatment.

WHAT IT ENCODES

When a myositis-SPECIFIC antibody (e.g., anti-HMGCR or anti-SRP for IMNM) is positive in the appropriate clinical and EMG/MRI context, muscle biopsy may be deferred and treatment started, since these antibodies are highly specific for their disease. This spares an invasive procedure in classic presentations.

BOARD TRAP

WRONG answer: extending 'biopsy not needed' to ANY positive antibody or an unclear picture. Discriminator: this shortcut applies only to highly specific antibodies in a clearly compatible presentation — in atypical, treatment-refractory, or antibody-negative cases (especially to exclude IBM, dystrophy, or mimics), biopsy is still required.

6. Anti-SRP aggressive

WHAT YOU SEE

The kneeling muscular figure labelled 'AGGRESSIVE / REFRACTORY' in the forge room — he embodies anti-SRP necrotizing myopathy: severe, rapidly progressive, steroid-resistant.

WHAT IT ENCODES

Anti-SRP (signal recognition particle) IMNM is the most AGGRESSIVE and treatment-REFRACTORY necrotizing subtype: very high CK, severe rapidly progressive proximal weakness, frequent dysphagia, and possible cardiac involvement. It usually requires combination immunosuppression upfront (steroids + IVIG + rituximab/methotrexate) and tends to respond more poorly than anti-HMGCR disease.

BOARD TRAP

WRONG answer: managing anti-SRP myopathy with corticosteroid monotherapy and expecting a quick response. Discriminator: anti-SRP IMNM is notoriously steroid-refractory and severe — it needs early aggressive combination therapy (often including IVIG and rituximab); steroids alone are inadequate.

7. IVIG

WHAT YOU SEE

The hanging IV bag labelled 'IVIG' with a 'stopping → improvement' cue — IVIG is a cornerstone/early agent for IMNM (especially anti-HMGCR), not a last resort.

WHAT IT ENCODES

IVIG is a cornerstone of IMNM treatment, particularly for anti-HMGCR disease where it is often used early and as monotherapy or add-on; it works rapidly and avoids the infection risk of heavy immunosuppression. Therapy should not be delayed because necrotizing myopathy can progress quickly to profound weakness. IVIG is also FDA-approved in dermatomyositis.

BOARD TRAP

WRONG answer: reserving IVIG only for refractory cases after prolonged steroid trials in IMNM. Discriminator: in anti-HMGCR necrotizing myopathy IVIG is a FIRST-LINE / early agent (sometimes alone), not a last resort — delaying it risks irreversible weakness.

8. Antisynthetase syndrome

WHAT YOU SEE

The banner reading 'ASyS' heading the right-hand panel — it opens the antisynthetase overlap room (myositis + ILD + mechanic's hands + Raynaud + arthritis + fever).

WHAT IT ENCODES

Antisynthetase syndrome (ASyS) is an overlap myositis defined by antibodies to aminoacyl-tRNA synthetases (most commonly anti-Jo-1). The clinical cluster: MYOSITIS + INTERSTITIAL LUNG DISEASE + mechanic's hands + Raynaud phenomenon + non-erosive arthritis + fever. ILD is frequently the dominant and prognosis-defining feature, and may even precede or occur without overt myositis.

BOARD TRAP

WRONG answer: classifying an ASyS patient as plain polymyositis and ignoring the lungs. Discriminator: the combination of myositis with ILD, mechanic's hands, Raynaud, arthritis, and fever defines antisynthetase syndrome — anti-synthetase antibody positivity mandates pulmonary evaluation (PFTs, HRCT), which routine PM workups omit.

9. Lungs decide prognosis

WHAT YOU SEE

The chest/lung sign reading 'LUNGS DECIDE PROGNOSIS' with 'ILD DOMINATES' on the right panel — interstitial lung disease, not the muscle, drives outcome in ASyS.

WHAT IT ENCODES

In antisynthetase syndrome, INTERSTITIAL LUNG DISEASE is the major determinant of morbidity and mortality, often outweighing the muscle disease. Evaluate with PFTs (watch FVC and DLCO) and high-resolution CT; ILD-predominant disease is treated with mycophenolate or rituximab plus steroids. Anti-MDA5 (an amyopathic-DM antibody) is the extreme — rapidly progressive ILD.

BOARD TRAP

WRONG answer: tracking only CK and muscle strength to gauge how an ASyS patient is doing. Discriminator: muscle improvement does not equal disease control — ILD drives prognosis, so serial PFTs (FVC/DLCO) and HRCT, not the CK, determine treatment intensity and outcome.

10. Mechanic's hands

WHAT YOU SEE

The cracked, fissured hands labelled 'MECHANIC'S HANDS' on the right — the dry split skin along the fingers, like a manual laborer's, is the ASyS skin clue.

WHAT IT ENCODES

Mechanic's hands = hyperkeratotic, cracked, fissured, hyperpigmented skin along the LATERAL/palmar surfaces of the fingers, resembling the hands of a manual laborer. It is a hallmark cutaneous sign of antisynthetase syndrome and overlaps with DM-spectrum skin disease; its presence should prompt antisynthetase antibody testing and ILD screening.

BOARD TRAP

WRONG answer: writing off cracked, fissured fingertips as occupational dermatitis or eczema. Discriminator: in a patient with myopathy, Raynaud, or arthritis, mechanic's hands is a marker of antisynthetase syndrome and points you toward anti-Jo-1 testing and HRCT for ILD, not topical steroids for dermatitis.

11. Anti-Jo1

WHAT YOU SEE

The 'Jo-1' antibody star badge marked 'MOST COMMON' on the right panel — the gold-star most-common antisynthetase antibody, the prototype for myositis + ILD.

WHAT IT ENCODES

Anti-Jo-1 (anti-histidyl-tRNA synthetase) is the MOST COMMON antisynthetase antibody (~20–25% of myositis, the majority of ASyS cases) and the prototype for myositis + ILD. Other antisynthetase antibodies (PL-7, PL-12, EJ, OJ, KS) cause the same syndrome; PL-7/PL-12 tend to have more lung and less muscle. Anti-Jo-1 positivity essentially guarantees ILD risk.

BOARD TRAP

WRONG answer: finding anti-Jo-1 and treating the muscle disease without pulmonary evaluation. Discriminator: anti-Jo-1 is an antisynthetase antibody tightly linked to interstitial lung disease — failure to screen with PFTs and HRCT (and to consider MMF/rituximab if ILD is present) is the classic miss.

12. Raynaud

WHAT YOU SEE

The hand drawn in tricolor white-blue-red labelled 'RAYNAUD' — the three-color digit change (ischemia→cyanosis→reperfusion) is part of the antisynthetase cluster.

WHAT IT ENCODES

Raynaud phenomenon — the triphasic white (ischemia) → blue (cyanosis) → red (reperfusion) color change of the digits with cold or stress — is part of the antisynthetase clinical cluster and of myositis overlap syndromes generally. It reflects the vasospastic/vasculopathic component and supports an overlap connective-tissue process rather than isolated myositis.

BOARD TRAP

WRONG answer: treating Raynaud in this setting as benign primary Raynaud needing no further evaluation. Discriminator: new Raynaud accompanying myositis, mechanic's hands, or arthritis suggests SECONDARY Raynaud within antisynthetase/overlap syndrome — it warrants antibody testing and ILD screening, not just hand-warming advice.

13. RTX / immunosuppression

WHAT YOU SEE

The shelf of drug bottles 'RTX', 'MMF', 'AZA' on the far right — the steroid-sparing immunosuppressants; RTX/MMF are favored when ILD is present (MTX is avoided).

WHAT IT ENCODES

Steroid-sparing immunosuppressants are added early in significant myositis: rituximab (RTX), mycophenolate mofetil (MMF), and azathioprine (AZA). For ILD-predominant antisynthetase or DM disease, MMF and RITUXIMAB are favored; methotrexate is generally AVOIDED when significant ILD is present (pulmonary toxicity risk). RTX is also useful in refractory IMNM and anti-synthetase disease.

BOARD TRAP

WRONG answer: reaching for methotrexate as the steroid-sparing agent in a patient with antisynthetase ILD. Discriminator: in the presence of interstitial lung disease, MTX is avoided because of pulmonary toxicity — MMF or rituximab is preferred; choose the steroid-sparing drug based on the lung status.

14. DM atrophy vs necrosis

WHAT YOU SEE

The biopsy panel top-right contrasting DM 'perifascicular fragmentation/atrophy' against scattered 'necrotic fibers' of IMNM — rim-of-fascicle atrophy = DM, scattered necrosis = IMNM.

WHAT IT ENCODES

Key biopsy contrast: PERIFASCICULAR ATROPHY (atrophic fibers at the rim of fascicles, from the microangiopathy/complement attack on perimysial capillaries) = dermatomyositis; SCATTERED MYOFIBER NECROSIS with minimal inflammation = immune-mediated necrotizing myopathy. Add the other two patterns for the full set: endomysial CD8+ invasion = PM; rimmed vacuoles = IBM.

BOARD TRAP

WRONG answer: equating any necrotic fibers on biopsy with DM, or any atrophy with IMNM. Discriminator: location and pattern matter — perifascicular (rim) atrophy is the DM fingerprint and abundant scattered necrosis with sparse inflammation is the IMNM fingerprint; mixing them up misclassifies the subtype and its antibody/treatment associations.
